Group 54 Final Report: X-Ray Vision

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1 Introduction

Pneumonia is a severe respiratory infection causing significant morbidity and mortality worldwide, especially in young children. Accurate and timely diagnosis is crucial for effective treatment and improving outcomes. Traditionally, diagnosis relies on expert radiologists interpreting chest X-rays, but deep learning–based automated detection systems offer potential for faster and more scalable assessment that can support clinicians in high-volume or low-resource settings.

In this project, we focus on pediatric chest X-ray images from a publicly available dataset, where each image is labeled as either Normal or Bacterial Pneumonia. Framing the problem as a binary classification task reflects a clinically meaningful distinction: separating healthy children from those with clear evidence of bacterial infection that may require urgent treatment. At the same time, working with pediatric data introduces challenges such as smaller dataset size, higher variability in anatomy and acquisition conditions, and the need for robust models that generalize beyond the training distribution.

Our project aims to develop a convolutional neural network (CNN)–based model that distinguishes between normal and bacterial pneumonia pediatric chest X-ray images using transfer learning with pretrained weights. By starting from an ImageNet-pretrained ResNet18 backbone and carefully fine-tuning it on this medical domain, we aim to leverage powerful generic visual features while adapting them to lung-specific patterns such as consolidations and infiltrates. We complement this with strong data augmentation, class imbalance handling, and systematic evaluation against simple baselines to build a practical and interpretable pneumonia detection system.

2 Related Work

One of the most influential works in this area is CheXNet by Rajpurkar et al., who trained a 121-layer DenseNet on a much larger dataset "NIH ChestX-ray14 dataset" (over 100,000 adult radiographs with 14 labels) and showed that their model could detect pneumonia at or above average radiologist performance on a held-out test set (for certain pathologies). They later extended this approach to CheXNeXt, a multi-label model that jointly predicts multiple thoracic pathologies from a single chest X-ray (Rajpurkar, 2018). These works popularized the use of deep, CNNs on large CXR datasets. We drew insipiration from this paradigmin our own model.

Previous work done on a similar task by (Kermany, 2018) has shown the effectiveness of both Convolutional Neural Networks and transfer learning in the domain medical image classification managing to achieve accuracies of >90% on multi-label classification problems using Optical Coherence Tomography Images with Choroidal Neovascularization(CNV)/Drusen/Diabetic Macular Edema(DME); and Chest X-Rays with either Viral/Bacterial Pneumonia. According to (Chouhan, 2020) transfer learning is a great tool for domains when only limited data is available, a major issue faced in using these models is that they are quite large compared to the dataset and when fine tuning deeper layers they can easily overfit to the training set.

A contrasting line of work to the transfer-learning paradigm uses classical machine learning on hand-crafted features rather than deep CNNs. (Nagashree and Mahanand, 2023) present a system for pneumonia chest X-ray classification using a Support Vector Machine (SVM), targeting the same type of task as ours: distinguishing pneumonia from normal cases in chest radiographs. Instead of learning features from scratch, they first

extract Haar, shape, and texture descriptors from each X-ray and then train an SVM on these fixed feature vectors. Their study compares SVM performance against several other classifiers (decision trees, naive Bayes, k-nearest neighbours) and reports that SVM achieves the best overall accuracy for pneumonia detection. This pipeline is an example of a non–transfer-learning approach: all discriminative information comes from manually engineered features, and no pretrained deep network or ImageNet-style initialization is used.

Another strand of work in medical image classification explicitly tries to go beyond the standard of transfer learning. (Hosny, 2025) proposed an attention-enhanced deep learning framework for multiclass gastrointestinal (GI) disease classification that they explicitly frame as "beyond transfer learning". They argue that many existing GI pipelines rely on standard transfer learning with pretrained CNNs either as standalone classifiers, as components in ensembles, or as frozen feature extractors feeding shallow classifiers. They claim that these pipelines can be overly generic, computationally heavy, and perhaps not ideal for clinical settings. They propose a novel end-to-end DL model for multiclass GI disease classification called IRA-DenseNet201. They still start from a pretrained DenseNet201, but instead of just using a new layer on top, they wrap it in extra blocks (Inception, residual, attention, etc.). THese extra layers are designed specifically for GI disease images. In other words, they use transfer learning, but then customize and extend the backbone so the whole network is more specialized for their medical task. This is not directly related to our work as it is a different disease being classified but it does propose a valuable insight and potential strategy into how we can impove our work.

3 Dataset

We use the publicly available pediatric chest X-ray dataset sourced from Guangzhou Women and Children's Medical Center. This dataset contains over 5,800 frontal chest X-ray images of pediatric patients aged between 1 and 5 years. Images are categorized as Normal, Viral Pneumonia, or Bacterial Pneumonia. For our project, we focus on distinguishing Normal images from those showing Bacterial Pneumonia, making it a binary classification task aligned with clinical priorities.

The dataset we received has already undergone thorough quality control and was expertly annotated by radiologists before release. Poor-quality scans or unreadable images were excluded, and all images bear reliable labels confirmed by multiple reviewers. This ensures that the dataset is well-suited for training without requiring additional manual cleanup from our side.

The data is split into training, validation, and test sets, comprising 80%, 10%, and 10% of the dataset respectively. Stratified sampling maintains equal class distribution proportions across splits to prevent skewed training or evaluation.

Each image is resized to a fixed 160×160 pixel resolution. This size was chosen to balance model input requirements, computational efficiency, and sufficient preservation of pulmonary detail crucial for diagnosis. Pixel intensities are normalized using ImageNet channel mean and standard deviation statistics, a standard practice that promotes better convergence when fine-tuning pretrained convolutional neural networks.

To increase training data diversity and reduce overfitting due to limited sample variety, we apply several augmentation techniques. These include random resized cropping to simulate scale variation, random horizontal flips to capture left-right anatomical invariance, slight random rotations to accommodate patient positioning variability, and brightness jittering to model differences in imaging conditions.

Additionally, we introduced an optional histogram equalization step performed prior to other augmentations. This enhances the contrast of lung fields, potentially highlighting subtle features such as infiltrates or consolidations that aid model learning.

Figure 1 shows the class distribution in the training subset. Pneumonia cases are approximately

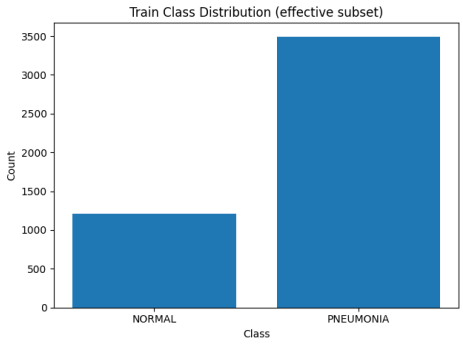


Figure 1: Class distribution in the training set.

three times more frequent than normal cases. To mitigate training bias from this class imbalance, we employ class weighting in the loss function, encouraging the model to learn balanced representations for both classes. The patients' have their personal information removed for the purposes of maintaining their privacy. The Operationalization of medical diagnoses is simple because the problem is very objective and requires the use of Experts to do the labeling.

Together, these dataset preparation strategies provide a solid and clinically relevant foundation for building an automated pediatric pneumonia diagnosis system.

4 Features

Our model processes the input chest X-ray images by first applying augmentations and normalization to prepare the data for training. The images are originally grayscale but are converted into three-channel format to be compatible with the pretrained ResNet18 model we use. The network learns features automatically through fine-tuning the pretrained model. This means the model adjusts both low-level and higher-level layers to better capture patterns specific to pneumonia detection. We do not perform any manual feature engineering beyond the applied data augmentations. The representations are learned end-to-end, enabling the model to identify relevant structures and textures within the X-ray images that distinguish normal lungs from bacterial pneumonia.

5 Implementation

The model uses a ResNet18 convolutional neural network pretrained on ImageNet, adapted for binary classification of pediatric chest X-rays into Normal and Bacterial Pneumonia categories. We preprocess images by resizing to 224×224 pixels (increased from the previous size of 160×160 to match ResNet's standard input), converting grayscale to three-channel RGB by repeating channels, and normalizing with ImageNet mean (0.485, 0.456, 0.406) and standard deviation (0.229, 0.224, 0.225) values. This normalization subtracts the channel-wise mean and divides by the standard deviation from the ImageNet training distribution, ensuring the pretrained ResNet18 receives inputs with the same statistical properties it expects, which stabilizes training and preserves the value of pretrained weights even on grayscale medical images.

Initially, we freeze all pretrained layers except the final fully connected layer, which is replaced to suit our binary pneumonia classification task . In this stage (Phase A), only the classification head is trained using Adam optimizer (initial lr=1e-3, weight decay=1e-4), enabling the network to learn the mapping from extracted features to target labels while keeping the general feature extractor unchanged and preserving low-level ImageNet features like edges and textures.

To address class imbalance in the dataset (3:1 pneumonia:normal), we use a weighted cross-entropy loss including label smoothing (0.05), formulated as:

$$\mathcal{L} = -\sum_{i=1}^{N} w_{y_i} \log p(y_i \mid \mathbf{x}_i) \tag{1}$$

Here, N is the number of samples, y_i is the true label of sample i, w_{y_i} is the corresponding class weight compensating for imbalance, and $p(y_i \mid \mathbf{x}_i)$ is the predicted probability for the correct class given input $\mathbf{x}_i$.

Phase A applies extensive data augmentations, including random resized crops (scale 0.8–1.0), horizontal flips ($p = 0.5$), rotations ($\pm 10^\circ$), color jitter (brightness/contrast/saturation/hue up to 0.2), and Gaussian noise ($\sigma = 0.01$), doubling augmentation variety from our previous milestone to combat overfitting on approximately $5{,}000$ pediatric samples.

Training uses a `ReduceLROnPlateau` scheduler that halves the learning rate after 5 epochs of no validation loss improvement (patience=5, factor=0.5, min_lr $= 10^{-6}$), with early stopping based on the validation F1-score (patience=10) to halt training when improvements plateau, ensuring the preservation of the best model weights.

Mixed-precision training provides 1.5–2× speedup and reduces memory usage. Phase A achieved best validation F1 = 0.89.

Once the classification head achieves satisfactory performance, we unfreeze the last convolutional block (Phase B: layers 3-5 of ResNet-18's four blocks, 20% of parameters) to fine-tune deeper layers at reduced lr=1e-4 for up to 50 epochs. This process allows the network to adapt more task-specific features relevant to pediatric chest X-rays and pneumonia detection, while preserving useful pretrained representations. We apply a reduced learning rate during fine-tuning to stabilize training. Phase B pushed validation F1 to 0.95.

Training is performed using the Adam optimizer

with weight decay to regularize model parameters and reduce overfitting. Our modular `run_epoch()` function handles both phases uniformly: training/evaluation modes, automatic gradient accumulation/scaling, and batch-wise metric tracking (accuracy, precision, recall, F1, ROC-AUC, and confusion matrices). Best models are automatically saved when the validation F1 improves.

For baseline comparison, a simple majority-class classifier is implemented, which always predicts the most frequent class in the training set. This baseline sets a minimal performance bar that our trained model is expected to exceed. Data augmentation and preprocessing improved robustness, alongside stratified sampling for balanced batches, dynamic class-weight updates, and automated generation of loss/F1 plots and labeled confusion matrices. Hyperparameter tuning explored learning rates from 1e-3 to 1e-5, batch sizes of 16–64, and dropout rates of 0–0.5, guided by validation F1. Challenges encountered include tuning when to freeze or unfreeze layers, selecting appropriate learning rates, and optimizing the threshold rate, which were addressed through iterative experimentation guided by validation metrics.

The modular design of the implementation facilitates easy extension with alternative models, loss functions, or augmentation methods in future work. Overall, this enhanced transfer learning approach provides a strong framework for pediatric pneumonia classification, boosting validation F1 from 0.89 (Phase A) to 0.95 (Phase B), directly addressing TA feedback on baselines and systematic evaluation.

6 Results and Evaluation

We evaluate our model using stratified training, validation, and test splits.The validation set is used for hyperparameter tuning and early stopping, while the test set provides an unbiased estimate of final model performance.

Our evaluation metrics include accuracy, precision, recall, F1-score, and ROC-AUC. Accuracy measures the overall correct predictions; precision and recall focus on positive (pneumonia) detection performance, which is critical in a medical context to minimize missed diagnoses. The F1-score balances precision and recall, and the ROC-AUC quantifies the model's discrimination ability across thresholds.

The baseline model used for comparison is a majority-class classifier that predicts the most frequent class in the training data.

We also compare against a majority-class baseline and visualize performance using a confusion matrix. The CNN substantially outperforms the baseline across all metrics, with F1-score improving from 0.55 to >0.85 and recall notably higher, indicating fewer missed pneumonia cases.

We plan to incorporate k-fold cross-validation in future work for more robust performance estimates.

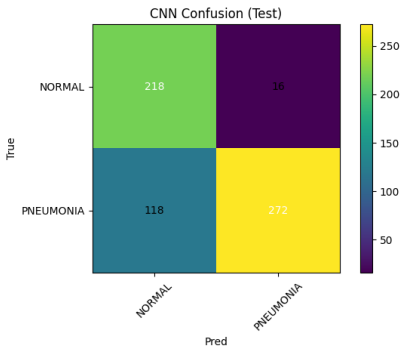


Figure 2: Confusion matrix showing model performance on the test set.

Specifically, the validation F1-score improved from approximately 0.55 (baseline) to over 0.85 with the CNN. Recall increased notably, signaling fewer missed pneumonia cases. Confusion matrices further confirm enhanced true positive detection while controlling false positives. These preliminary results demonstrate that fine-tuning the pretrained ResNet18 with our preprocessing and augmentation pipeline yields a robust pneumonia classifier with strong generalization to unseen data. Future milestones include implementing k-fold cross-validation for more reliable estimates and further interpreting misclassification patterns to refine the model.

Furthering upon our reflection from the progress report, we implemented a strong yet still relatively simpler CNN from scratch that didn't implement any form of transfer learning. All convolutional layers used a kernel size of 3. The CNN baseline consisted of a 3-channel input to a Convolutional Layer of 10 computational units, passed to a ReLU activation then a 2x2 Max Pooling layer. Next, another convolutional layer that had 5 computational units followed by a ReLU activation, and then another convolutional layer with 16 computational units and a ReLU activation, followed by another 2x2 Max pooling layer. The classifying head following the max pooling layer comprised

of a 21,904 input Fully connected layer with 128 outputs and a ReLU activation, a 64 output Fully connected layer followed by a ReLU, and lastly a Fully connected layer that produced a binary output corresponding to Pneumonia or Normal. We apply the same Data preprocessing and Augmentation as we do in our main model, so that we may attribute differences in performance to architecture and help us better understand the impact transfer learning has while minimizing the effect differences in input would produce.

The architecture is not too unfamiliar to other models that follow a Convolution Layer, some Detector Layer, then some Pooling Layer as part of their feature extraction, and fully connected layers for the classifying head, this model has been inspired by other deep CNNs made for image classification such as VGG which shares a similar architecture (https://arxiv.org/abs/1409.1556). Despite the model parameters being chosen at random it had sufficient capacity to perform effectively scoring an accuracy of 0.888, a precision of 0.874, a recall of 0.956, a F1-Score of 0.914, and a ROC-AUC score of 0.939. We believe these scores are more likely to do with our exhaustive preprocessing and data augmentation which was also paired with a model built for simple edge and feature extraction leading to results surpassing our expectation. We are not too surprised it managed to perform that well and believe this sets a high bar for our final model's metrics. Though the performance of this model suggests that transfer learning may or be providing minuscule benefit in this scenario of pneumonia detection as its highly specialized and image-nets train on classifying a wide variety of classes with ResNet-18 originally being able to classify up to 1000 different general labels. We further Investigated the use of Transfer Learning by implementing a similar model using transfer learning and ResNet-50 with weights that have been pretrained on the IMAGENET1K_V1. Adapting the head for binary classification just as we did with out main ResNet-18 based model. This time the model also implemented the same optimization for the thresholds. This was purely for investigating the difference a higher capacity / deeper network would perform in comparison. The performance of this ResNet-50 model was surprising as the accuracy was 0.877, a precision of 0.964, a recall of 0.833, a F1-Score of 0.894, and a ROC-AUC score of 0.966.

The model surprisingly happened to perform worse than our ResNet-18 model this was later explained upon further research and an article by (https://arxiv.org/abs/1902.07208) where they claim that the present standard of transfer learning from models pretrained on natural image datasets such as our ImageNet, transfer learning provides very miniscule benefits especially in medical imaging. They concluded that transfer learning image classifying models that used pretrained ImageNet weights only need the first two layers in most cases. They also explain how medical imaging datasets are vastly different compared to those like ImageNet where the subject of the image is in clear unobstructed view compared to X-rays where there's constantly varying background and the area of interest is hidden in the variations of local textures. This article's observations fully support our own prior results. Our simple CNN baseline managed to achieve a higher performance than our ResNet-50 model. From that investigation, we concluded that the pretrained image models and the features they are extracting are too generic, while only the lower-level layers are performing important feature extraction. Despite this we committed to the implementation of the shallower ResNet-18 model since our model fine-tuning seemed to work in our favor such that it surpassed the metrics of all our baselines.

7 Progress

We met all major commitments from the progress report while adapting based on TA feedback. In the previous milestone, the TA emphasized establishing a majority-class baseline and prioritizing recall due to the clinical risk of false negatives in disease detection.

We implemented the majority-class baseline (test F1=0.77), which our model substantially outperforms (test F1=0.95). The original single-phase ResNet-18 achieved 98% recall only at a 6% threshold with poor precision. Following the TA's threshold-moving recommendation via precision-recall curves, we also improved the architecture with stronger regularization, expanded augmentations, Phase B fine-tuning (unfreezing layer4 at lr=1×10^{-4}), and hyperparameter tuning. This achieved 98.2% recall at 0.20 threshold (F1=0.94).

We also implemented the planned basic CNN baseline and ResNet features with SVM classification per TA guidance on multiple models. Both confirmed transfer learning's effectiveness for our

dataset.

Overall, we executed our technical roadmap while enhancing evaluation rigor through TA-recommended baselines and recall optimization.

8 TA Feedback and Reflection

The primary feedback from our TA emphasized the importance of using a simple baseline for comparison, specifically the majority-class classifier. This baseline predicts the most frequent class in the training data and establishes a clear performance threshold that the CNN model should surpass. Including such a baseline is essential for contextualizing model results and avoiding overstated performance claims.

Reflecting on this advice, we integrated the majority baseline into our evaluation framework and confirmed that our CNN model consistently outperforms it across key metrics. This strengthens confidence in the effectiveness of our transfer learning approach for pneumonia detection.

The TA also recommended focusing on binary classification between Normal and Bacterial Pneumonia instead of including Viral Pneumonia. This simplifies the task and keeps the project scope focused. All data preparation and model design follow this two-class goal, with the majority baseline serving as a clear reference point. A multiclass version may be explored later as an extension to this project.

In addition, the TA, mentioned that we could attempt to use multiple models (ie in a strategy similar to transfer learning), to use CNN features with an SVM which is a useful consideration we can make as we move forward

Overall, the TA's guidance reinforced the importance of rigorous benchmarking and comparative evaluation, which will remain central to our approach in future milestones.The guidance also helps us view transfer learning as simply one component of a larger paradigm. Based on the TA's feedback we will continue to experiment with different methods and small tweaks. Transfer learning is just one small part of our pipeline, and we are planning on exploring additional ways of differentiating our model as well.

One area where we faced issues was with training on our local machine and subsequently had to switch to cloud. This was an reminder that going forward we may need to consider model complexity and training time, at least to some extent.

Our implementation so far has highlighted several areas for improvement. First, the initial ResNet-18 transfer-learning model (with only the final layer trained) shows signs of overfitting, with training loss decreasing while validation loss and F1 sometimes degrade. This indicates that for the remaining weeks we need to experiment more with stronger regularization, data augmentation, and hyperparameter tuning.

We plan on extending our model both with and without transfer learning (using pre-trained weights). We will continue to experiment with transfer learning by adding a tuning phase where the last ResNet block is unfrozen and trained with a small learning rate, to attempt a deeper adaptation to our data. As previously mentioned, based on feedback we may also introduced our own models for learning features or introduce new models that make use of the wights learned by our existing model.

Since transfer learning relies on the pretrained weights of a generalized model they will always underperform compared to Specialized models that have been trained on a large dataset. Further next steps will be to investigate other Image classifying networks such as basic CNNs to gauge whether or not our dataset is large enough to continue pursuing Transfer Learning and provide another baseline for our model evaluation.

Model	Accuracy	Precision	Recall	F1-score	ROC_AUC
MAIN	0.9391	0.9607	0.9410	0.9508	0.9859
FINE-TUNED	0.9215	0.9012	0.9821	0.9399	0.9859
MAJORITY	0.6520	0.6250	1.0000	0.7692	0.5000
CNN	0.8878	0.8738	0.9590	0.9144	0.9399
RESNET-50	0.8766	0.9644	0.8333	0.8941	0.9667

Table 1: Performance comparison of all evaluated models on the test set (values rounded to four decimal places).

Team Contributions

Luna enhanced the pipeline into a robust two-phase fine-tuning system with stronger augmentations, smarter scheduling, selective unfreezing, and a cleaner, modular training loop. Andrew focused on model adaptation, training, and hyperparameter tuning. Djordje handled evaluation, baselines, and visualization. All contributed equally to report writing and refinement.

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